Game Theory and Social Choice

Lecture 1: Course Overview

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Introduction

Two Questions

This course is organised around two questions.

1. What is rational choice when the results of your choice also depend on the choices of other rational agents?
2. What does it mean for a group to choose?

We will spend roughly half the semester on each.

Game Theory

Question One

What is rational choice when the results of your choice also depend on the choices of other rational agents?

In standard decision theory, the things you are uncertain about are passive. So here's the basic kind of choice you discuss

Table 1: A decision that rests on 'passive' facts

Choice	Rain	Dry
Take Umbrella	1	2
Don't Take Umbrella	0	3

The states of the world, the things represented on the columns, do not react to, or anticipate, your choices.

Question One

But many of the choices we care about involve other people, who are themselves trying to choose well, and whose choices depend on what they think you will do. In cases of repeated choice, they may **react** to our earlier choice.

A Coordination Problem

Two friends agree to meet at a café in Ann Arbor but did not say which. Both prefer Comet to Lab, and both prefer meeting somewhere to not meeting.

Table 2: A simple coordination game

	Comet	**Lab**
Comet	4, 4	0, 0
Lab	0, 0	3, 3

What should each friend do?

What Makes It Hard

Comet looks like the right answer. They get a higher payoff if they coordinate there.

But if each were certain that the other would choose Lab, then it would make sense to choose Lab.

In general, what choices we make depends on what we think other people, who are facing a similar choice, will do.

An Even Harder Case

Maybe you think that case is easy; they'll surely coordinate on the better option. But there are harder variants of the case.

Imagine that while they want to coordinate, they want to get some work done if they don't coordinate, and they find Lab better for working in alone. Then the payoffs might look like this.

Table 3: A not so simple coordination game

	Comet	**Lab**
Comet	4, 4	0, 2
Lab	2, 0	3, 3

As we'll see when we get further into the course, there is no scholarly consensus on what choice is rational in this case, or even if there is a uniquely right answer.

The Prisoner’s Dilemma

These strategic choices can lead to distinctive puzzles; this might be the most famous one.

Table 4: A canonical Prisoner’s Dilemma

	Cooperate	**Defect**
Cooperate	3, 3	0, 5
Defect	5, 0	1, 1

Each player has a dominant strategy: defect, regardless of what the other does.

So two rational players both defect. Both get 1, when both could have gotten 3.

Why It Is a Dilemma

There is a sense in which the players have done what each had reason to do.

There is also a sense in which they have collectively failed.

There are two important takeaways from this.

1. The simple question “What is it rational to do?” is hard even in really simple cases.
2. It’s really important to keep distinct, at least conceptually, what’s good for the group, and what’s good for each member of the group.

What the Game-Theory Half Will Do

In regular decision theory there is one canonical solution concept: Maximise Expected Utility.

In game theory, we work through several distinct solution concepts, and (a) what relations hold between them, and (b) which of them is best suited to different problems.

Some of this profusion arises in games like the ones we've seen so far: games where the players move once, simultaneously.

But once we add time into our games, there are even more options.

The most complicated notions we'll look at are designed primarily to deal with *signalling* games, and include things that we often slide over in formal models in philosophy, like how to update on probability 0 events.

Social Welfare

Question Two

What does it mean for a group to choose?

Different individuals have different preferences. Can we combine those preferences into a single group preference? When can we say what the group has decided?

The primary puzzle here concerns aggregation rather than strategy, though strategic considerations will never totally go away.

As you might already be familiar with from thinking about elections, or about everyday interactions, the choice of an aggregation procedure needs to be sensitive to the strategic behaviour of group members.

Condorcet's Paradox

Here's a simple case that will already be enough to get some problems going. This clean formulation comes from the 18th century theorist and radical politician, the Marquis de Condorcet.

There are three (numbered) voters, and three (lettered) options. Each column is a voter's ranking, top to bottom.

Table 5: Condorcet's Paradox

Voter 1	Voter 2	Voter 3
A	B	C
B	C	A
C	A	B

Condorcet’s Paradox

Voter 1	**Voter 2**	**Voter 3**
A	B	C
B	C	A
C	A	B

Let’s say we adopt the rule that the group prefers X to Y when a majority of its members do. Then we get that the group prefers A to B, and prefers B to C, and prefers C to A. Oops!

Generalizations

Condorcet noticed this in the 1780s.

Kenneth Arrow (1951) generalised this to show that no method of aggregating preferences can satisfy a small set of attractive conditions.

Amartya Sen (1970) showed that you cannot combine a minimal commitment to individual liberty with even some principles about preference aggregation that looked beyond serious dispute.

The main plotline here is that we'll keep seeing **impossibility theorems**, results that say some things which we really wanted our aggregation procedures to have can't all obtain. So what do we do?

What the Social-Choice Half Will Do

We work through Amartya Sen's *Collective Choice and Social Welfare* (originally published in 1970, with an expanded edition in 2017 that we'll work through).

Obviously there are lots of great economists around, and lots of great philosophers, but you'd be hard pressed to find someone who is better than Sen at *both*. He's also an excellent writer, though this doesn't *always* come through in CCSW.

Some of the topics we'll cover include:

- Arrow's Theorem (including Sen's preferred proof of it)
- The Paretian liberal paradox that Sen made famous
- Issues about comparing interpersonal welfare
- Issues about the relationship between welfare, equity, and justice
- Sen's preferred theory of welfare: the **capability** approach.

Overview

The Unifying Theme

The two questions look different. But there is a common theme.

> **Note**
>
> Individual rationality does not extend cleanly to social settings.

Game theory and social choice theory are two different ways of seeing how, and what philosophers have made of it.

Course Shape

Through the midterm break, we'll primarily be doing game theory.

After that, we'll focus on social choice theory, primarily working through Sen's book.

Mathematical Level

This course is more mathematical than the average philosophy course, though less mathematical than many (I guess most) economics courses.

The mathematics will be at roughly the level of formal work in a philosophy journal. You should expect sets, functions, probabilities, expected utilities, and some algebra. **Calculus** will not be needed.

If you have taken a philosophy course in formal methods, you'll be fine. Similarly, if you've done any stats or econ courses, you'll be fine.

If you have not, **ask questions**; the formalism is there to make the structure visible.

Required Reading

One required book: Amartya Sen, *Collective Choice and Social Welfare: An Expanded Edition* (Harvard University Press, 2017). Available electronically through the U-M library.

Game theory readings will be a mix of lecture notes and papers/chapters available through Canvas

Assessment

Component	Weight
Two essays (25% each)	50%
Eight quizzes (best six count)	40%
Class participation	10%

The quizzes are going to be relatively technical; the essays will be more discursive.

Practicalities

- Lectures: Tuesday and Thursday, [time and room]
- Office hours: to be announced
- Course website: bweatherson.github.io/F26-Phil444-site
- Canvas: readings, assignments, announcements

Next Lecture

On Thursday we'll do a flying recap of decision theory under uncertainty, along with a tour of some of the famous problems.